Immunology and Inflammation

Susceptibility of *Kit*-mutant mice to sepsis caused by enteral dysbiosis, not mast cell deficiency

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eLife Assessment

This study presents a **useful** finding showing that the high susceptibility to sepsis of Kit-mutant mice is due to dysbiosis. However, the data provided is **incomplete** and would benefit from more rigorous approaches. With the mechanism part strengthened, this paper would be of interest to researchers on mast cell biology and mucosal immunology.

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Abstract

Kit-mutant mice are highly susceptible to polymicrobial sepsis elicited by cecal ligation and puncture (CLP). This vulnerability has been attributed to the mast cell deficiency of *Kit* mutants, suggesting important roles of mast cells in defense against bacteria. We show here that mice lacking mast cells but wild-type for *Kit* are as resistant to sepsis as mast cell-proficient mice, excluding mast cells as protective factor. Induction of sepsis by direct injection of intestinal microbiota instead of surgical gut perforation revealed comparable protection of *Kit*-deficient and *Kit* wild-type mice, indicating normal bacterial immune defense in the absence of Kit. Notably, compared to wild-type mice, we observed more that 1000-fold greater *E. coli* colony-forming units in the cecal content of *Kit*-mutant mice, consistent with dysbiosis from gastrointestinal pathophysiology. Thus, upon intestinal puncture, this vast overrepresentation of pathogenic bacteria led to incomparable infections, likely explaining the apparent susceptibility of *Kit*-mutants. These findings highlight the importance of considering potential effects of genetic mutations on endogenous microbiota composition in cecal ligation and puncture studies of mutant mice. Collectively, our results suggest that the susceptibility of *Kit*-mutant mice to sepsis is associated with their enteral dysbiosis rather than mast cell-deficiency.

Introduction

Nearly 30 years ago, two hallmark papers studying acute peritoneal infections reported evidence for protective functions of mast cells in anti-bacterial defense[1,2]. These experiments were performed in mice carrying inactivating mutations in the gene encoding the receptor tyrosine kinase Kit (*Kit*$^{W/Wv}$ mice) which, at the time, were the widely used model of mast cell deficiency owing to the dependency of mast cell development on Kit expression[3]. Both reports showed that *Kit*$^{W/Wv}$ mice were highly susceptible to peritoneal infections compared with *Kit* wild-type mice. Based on mast cell reconstitution and investigation of associated cytokines, the central conclusion was that the absence of mast cells in *Kit*$^{W/Wv}$ mice, and in particular the absence of mast cell-derived tumor necrosis factor alfa (TNFα), was responsible for the observed immunodeficiency[1,2]. Subsequently, mast cells were considered as key effectors in bacterial defense[4].

At the time, it was well established that *Kit* mutations affect many cell lineages and tissues in developing and adult mice beyond the defect in mast cells, and hence it remained ambiguous whether or not a phenotype observed in *Kit*-mutant mice was indeed due to the absence of mast cells. With the advent of *Kit* mutation-independent mouse models of mast cell deficiency[5–8], which are by comparison to *Kit*-mutants highly specific experimental models[9–13], mast cell functions can be probed more conclusively in vivo. It has since become possible to independently confirm, or disprove, what had been concluded earlier on mast cell physiology based on work in *Kit*-mutants. Around 15 years into this process of re-evaluation of mast cell functions, a long list of suggested functions have not been reproduced when revisited in *Kit*-independent models (for discussions and references, see[9–13]).

Here, we have re-addressed the question whether mast cells are involved in immunity to acute peritonitis and sepsis. We confirmed the susceptibility of *Kit*-mutant mice (*Kit*$^{W/Wv}$) to sepsis[1] in the cecal ligation and puncture (CLP) model[14]. However, comparison of *Kit* wild-type mice with (*Cpa3*$^{+/+}$) or without (*Cpa3*$^{Cre/+}$) mast cells[6] showed that the outcome of such infections was independent of mast cells, i.e. mast cells were not required for resistance to polymicrobial sepsis in the CLP assay. Furthermore, *Kit*$^{W/Wv}$ mice exhibited the same susceptibility as *Kit*$^{+/+}$ mice upon direct injection of intestinal bacteria, pointing at a defect in *Kit*$^{W/Wv}$ mice related to the intestinal injury or the release of endogenous flora from the cecum in the CLP model. Finally, we identified markedly higher *Escherichia coli* (*E. coli*) colony forming units in the intestines of *Kit*$^{W/Wv}$ mice compared to *Kit*$^{+/+}$ mice. The release of higher numbers of *E. coli* from the *Kit*$^{W/Wv}$ mouse intestine during CLP sepsis induced a more lethal infection. Accordingly, co-housing neutralized the susceptibility of *Kit*-mutant mice to CLP-induced sepsis. Our data suggest that the dysbiosis of *Kit*$^{W/Wv}$ mutants has been misinterpreted as immunodeficiency. Collectively, we present evidence that sepsis susceptibility of *Kit*-mutant mice is due to elevated microbiota pathogenicity, whereas mast cells do not play a role in protecting against polymicrobial sepsis.

Results

Mast cells are dispensable for survival of cecal ligation and puncture-induced polymicrobial sepsis

We subjected mast cell-deficient *Cpa3*$^{Cre/+}$ mice[6] on the C57BL/6 background (*Cpa3*$^{Cre/+}$), and their wild-type littermates (*Cpa3*$^{+/+}$) to the surgical model of cecal ligation and puncture[14]. Under mild sepsis conditions, induced by a 50 % ligation of the cecum (tying at the middle between the base and the pole of the cecum) and a single puncture of the cecum with a 25-gauge (25 G) needle, all wild-type mice (*Cpa3*$^{+/+}$) survived (**Fig. 1A**). Interestingly, all mast cell-deficient *Cpa3*$^{Cre/+}$ mice also survived (**Fig. 1A**). To explore whether a role for mast cell would become evident under more severe conditions we induced CLP using a 22 G needle (**Fig. 1B**). The survival (Methods) within two weeks after surgical puncture dropped to about half of the animals for both the *Cpa3*$^{+/+}$ and *Cpa3*$^{Cre/+}$ mice (**Fig. 1B**). We analyzed large cohorts of about n = 50 for each genotype, and found no statistically significant difference between both groups (p = 0.23), indicating that the mortality in cecal ligation and puncture sepsis was independent of mast cells. These results were surprising given that previous publications using *Kit*-mutant, mast cell-deficient mice concluded that there was a fundamental requirement for mast cells in sepsis survival[1,2]. As a control, we therefore included *Kit*$^{W/Wv}$ mice in our studies. Even under mild sepsis conditions, which all *Kit* wild-type (*Cpa3*+/+ or *Cpa3*$^{Cre/+}$) mice survived, 75 % of *Kit*$^{W/Wv}$ mice died (**Fig. 1A**), and in severe sepsis all *Kit*$^{W/Wv}$ mice succumbed the treatment within two days (**Fig. 1B**). *KitW/Wv* mice are F1 offspring from WB-*Kit*$^{W/+}$ and B6-*Kit*$^{Wv/+}$ parents, hence they have a genetic F1 hybrid (WBB6F1) background[15] (we refer here to WBB6F1-*Kit*$^{W/Wv}$ mice as *Kit*$^{W/Wv}$ mice). To exclude genetic background differences as a cause of the higher susceptibility of *Kit*$^{W/Wv}$ mice compared to *Cpa3*$^{Cre/+}$ mice, we repeated the CLP experiments using all mice on the WBB6F1 background (WBB6F1-*Cpa3*$^{+/+}$, WBB6F1-*Cpa3*$^{Cre/+}$ and *Kit*$^{W/Wv}$). On this background, mast cell-proficient WBB6F1-*Cpa3*$^{+/+}$ and mast cell-deficient WBB6F1-*Cpa3*$^{Cre/+}$ mice were both even more resistant to CLP than on the B6 background (**Fig. 1C**), which is consistent with an earlier report[16], and hybrid vigor. In total, 83 % of WBB6F1-*Cpa3*$^{+/+}$ and 74 % of WBB6F1-*Cpa3*$^{Cre/+}$ mice survived (p = 0.24), while again all *KitW/Wv* mice succumbed to sepsis. These experiments demonstrate that mast cells are not involved in sepsis resistance, while *Kit*-mutants are highly susceptible.

Because absence of mast cells in *Kit*$^{W/Wv}$ mice has been implicated in impaired cytokine responses in sepsis, notably in TNFα[1,2], we next analyzed the inflammatory cytokine response. Serum levels of TNFα, IL-6, and MCP-1 were measured at three different time points after CLP (**Fig. 1D-F**). Mast cell-proficient *Cpa3*$^{+/+}$ and mast cell-deficient *Cpa3*$^{Cre/+}$ mice showed very similar time course and amounts of cytokine production. The serum levels for each of the three cytokines was maximal after 8 h, and declined until 24 h without any significant further change at 48 h after cecal ligation and puncture. In contrast, *Kit*$^{W/Wv}$ mice started at 8 h with cytokine levels similar to *Cpa3*$^{+/+}$ and *Cpa3*$^{Cre/+}$ mice, but mounted an almost 10-fold higher TNFα, IL-6 and MCP-1 response at 24 h after CLP (**Fig. 1D-F**). Sham-treated animals (dashed lines) had transiently increased levels of cytokines at 8 h, in response to the anesthesia and laparotomy (**Fig. 1D-F**). In **Fig. 1G-I** we plotted the 24 h serum levels of individual mice, comparing *Cpa3*$^{+/+}$, *Cpa3Cre/*+ and *Kit*$^{W/Wv}$ mice. Median cytokine levels were similar in *Cpa3*+/+ and *Cpa3*$^{Cre/+}$ mice but elevated in *Kit*$^{W/Wv}$ mice. Comparable cytokine responses in *Cpa3*$^{+/+}$ and *Cpa3Cre/*+ mice are consistent with their similar survival rates. Clearly, and in contrast to earlier reports[1,2], the TNFα response was independent of the presence of mast cells, excluding these cells as the main source for TNFα in experimental peritoneal sepsis. Elevate cytokine responses were associated with the *Kit* mutation. Rather than protection, the strongly elevated inflammatory cytokine levels 24 h after cecal ligation and puncture in most *Kit*$^{W/Wv}$, and in some *Cpa3*$^{+/+}$ and *Cpa3*$^{Cre/+}$ mice may reflect an exaggerated immune response before succumbing to sepsis.

Collectively, these experiments demonstrate that mast cells do not play a role in protecting against enteric bacterial sepsis in the CLP model. We also confirmed the severely impaired survival of *Kit*$^{W/Wv}$ mice, and found that this was unrelated to deficiencies in production of TNFα, IL-6, or MCP-1.

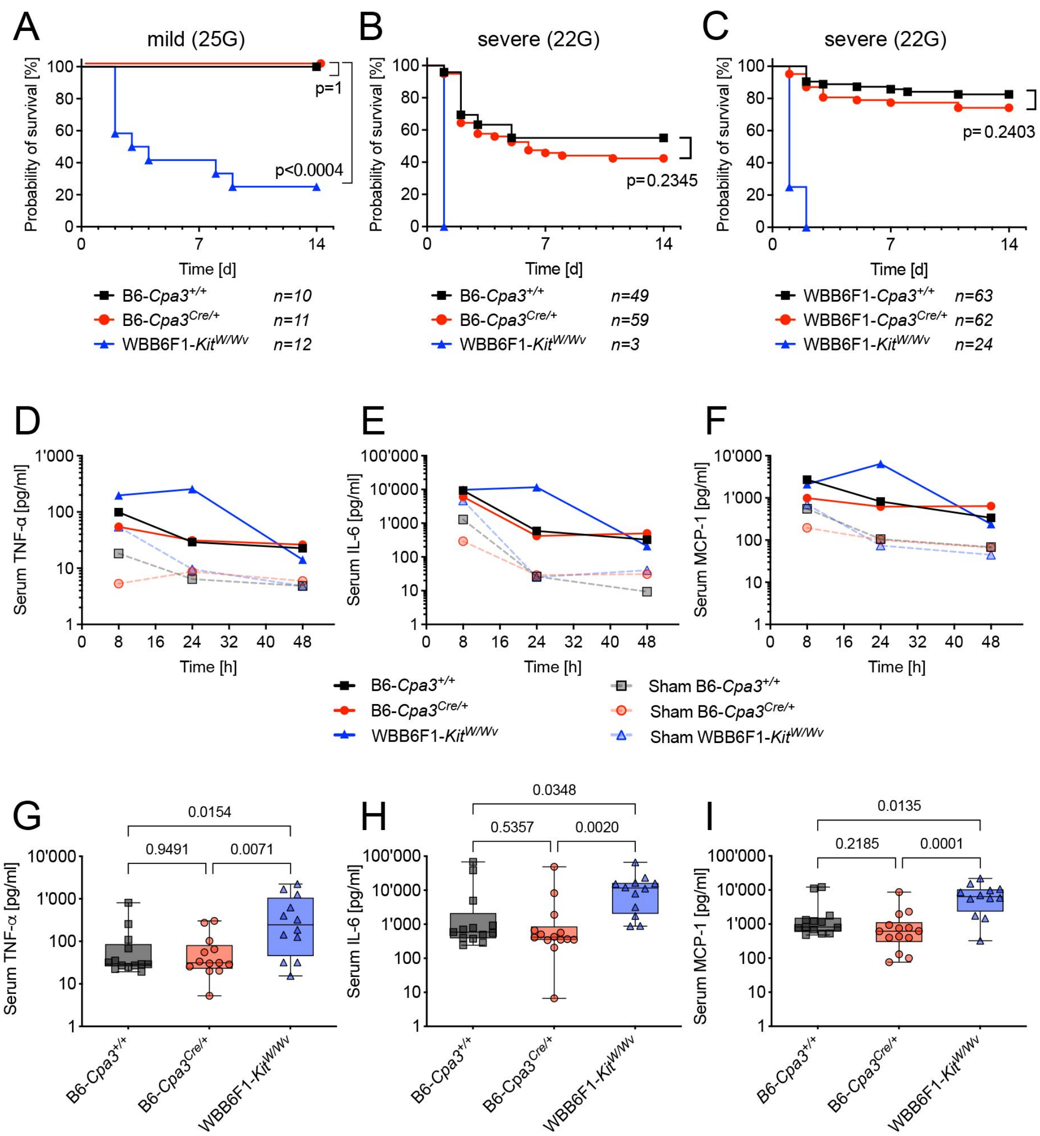

Figure 1.

$Kit^{W/Wv}$ mutant mice, but not mast cell-deficient $Cpa3^{Cre/+}$ mice, are susceptible for cecal ligation and puncture sepsis.

Survival of B6-$Cpa3^{+/+}$ (100% of n = 10), B6-$Cpa3^{Cre/+}$ (100% of n = 11), and WBB6F1-$Kit^{W/Wv}$ (25 % of n = 12) mice to mild (25 G needle) cecal ligation and puncture. WBB6F1-$Kit^{W/Wv}$ mice vs. B6-$Cpa3^{+/+}$ ($p < 0.0004$).

(B). Survival of B6-$Cpa3^{+/+}$ (55 % of n = 49), B6-$Cpa3^{Cre/+}$ (42 % of n = 59), and WBB6F1-$Kit^{W/Wv}$ (0 % of n = 3) mice under severe (22 G needle) cecal ligation and puncture. B6-$Cpa3^{+/+}$ vs. B6-$Cpa3^{Cre/}$ p = 0.2345. WBB6F1-$Kit^{W/Wv}$ vs. B6-$Cpa3^{+/+}$ $p < 0.0001$.

(C). Survival of WBB6F1-$Cpa3^{+/+}$ (82.5 %, n = 63), WBB6F1-$Cpa3^{Cre/+}$ (74.2 %, n = 62), and WBB6F1-$Kit^{W/Wv}$ (0 %, n = 24) mice to severe (22 G needle) cecal ligation and puncture. P-values for curve comparisons in A-C were calculated using the Mantel-Cox Log-rank test.

(D-F). Kinetics (in hours) of median serum concentrations for TNFα (D), IL-6 (E), and MCP-1 (F) in B6-$Cpa3^{+/+}$, B6-$Cpa3^{Cre/+}$, and WBB6F1-$Kit^{W/Wv}$ mice after severe (22-G needle) cecal ligation and puncture (solid lines), or after sham operations (same procedure but without cecal ligation and puncture) (dashed lines).

(G-I). From the kinetic shown in D-F, box plots were drawn for the 24-h time point indicating serum concentrations for TNFα, IL-6, and MCP-1 in B6-$Cpa3^{+/+}$, B6-$Cpa3^{Cre/+}$ and WBB6F1-$Kit^{W/Wv}$ mice. Each dot represents an individual mouse. The boxes extend from the 25th to 75th percentiles and the median is indicated. Whiskers range from minimum to maximum. P-values were calculated on log-transformed values by one-way ANOVA with Tukey's correction for multiple comparisons.

Mast cell-deficient $Kit^{W/Wv}$ mice resist intraperitoneal injection with cecal bacteria

Kit is essential for the development of interstitial cells of Cajal and for intestinal pacemaker activity, and hence autonomic gut motility is largely abrogated in *Kit*-mutants[17,18]. Given this dysfunction of the intestinal physiology in $Kit^{W/Wv}$ mice, we considered the possibility that the surgical CLP procedure, involving tissue ligation and perforation, could contribute to the sensitivity of $Kit^{W/Wv}$ mice to sepsis. We therefore induced sepsis without injury of the colon by injection of a cecal bacterial slurry containing a defined amount of enteral bacteria. This is a reproducible, bacteria dose-controlled peritoneal infection model[19]. Donors for enteral slurry were C57BL/6 mice. More than 90 % of all three genotypes ($Kit^{W/Wv}$, $Cpa3^{+/+}$ and $Cpa3^{Cre/+}$ mice) survived injection of low dose (3×10^8) bacteria (**Fig. 2A**). Survival was reduced to less than 50 % during the observation period of 10 days at an approximately 5-fold higher dose (14×10^8 bacteria) (**Fig. 2B**). For both doses, the outcome for *KitW/Wv* mice was statistically comparable to that of wild-type control mice (**Fig. 2A, B**), ruling out immunodeficiency to explain the sensitivity of $Kit^{W/Wv}$ mice in cecal ligation and puncture. Mast cell-deficient $Cpa3^{Cre/+}$ mice were no more susceptible than their mast cell bearing littermates (**Fig. 2A, B**), confirming the irrelevance of mast cells in this bacterial injection model.

In order to measure early-stage inflammatory cytokine release, low dose cecal slurry infection was repeated, and mice were analyzed one or two hours later. Compared to NaCl control injected animals, TNFα, IL-6, MCP-1 and IL-10 were elevated in the peritoneal lavage already after one hour, and the cytokine concentrations further increased at two hours after bacteria injection, in particular for IL-6 and MCP-1 (**Fig. 2C-F**). We could not detect increases in IFN-γ or IL12p70 (data not shown). Comparison of $Cpa3^{+/+}$, $Cpa3^{Cre/+}$ and $Kit^{W/Wv}$ mice revealed no significant difference in the cytokine responses, thus excluding mast cells as a major source of inflammatory cytokines, and indicating that this cytokine response is independent of Kit. Since mast cells have been implicated in the recruitment of innate immune cells, we measured neutrophils and macrophages in the peritoneal lavage fluid by flow cytometry one and two hours after bacterial injection (**Fig. 2G+H**). Neutrophils which are normally absent, or present only at very low numbers in the peritoneal cavity of naïve mice, were rapidly infiltrating the peritoneal cavity of infected animals (**Fig. 2G**). Of note, we counted very similar numbers of neutrophils (Gr1+CD11b+) in mast cell-proficient *Cpa3+/+*, and mast cell-deficient *Cpa3Cre/+* mice. Neutrophil recruitment was slightly reduced in $Kit^{W/Wv}$ mice which could be due to their global Kit-dependent neutropenia[20,21]. Numbers of macrophages (F4/80+CD11b+) declined[22] equivalently within the first hour of infection in $Cpa3^{+/+}$, $Cpa3^{Cre/+}$ and $^{KitW/Wv}$ mice (**Fig. 2H**). Taken together, in contrast to the cecal ligation and puncture sepsis results, $Kit^{W/Wv}$ mice show normal survival, cytokine response and neutrophil recruitment after bacterial slurry injection. This implies that the susceptibility of $Kit^{W/Wv}$ mice to cecal ligation and puncture sepsis does not reflect defects in immunity, and that immunity is again unaffected by the absence of mast cells.

Moreover, our time course experiments show that, within the first hours of peritoneal bacterial infection, mast cells are not major producers of inflammatory cytokines.

$Kit^{W/Wv}$ mice harbor intestinal microflora with increased pathogenic potential

In the CLP model, sepsis is driven by the mouse's own intestinal flora. The selective susceptibility of $Kit^{W/Wv}$ mice, but not $Cpa3^{Cre/+}$ mice, in this model raises the question whether impaired intestinal physiology, i.e. hypomotility of gut peristalsis and impaired gastrointestinal transit observed in *Kit*-mutants[18,23], may influence the composition and pathogenicity of the intestinal microflora. To directly compare the pathogenicity, we isolated enteral bacteria from either $Kit^{+/+}$; or $Kit^{W/Wv}$ mice (donor flora) and injected them into recipient cohorts of $Cpa3^{+/+}$

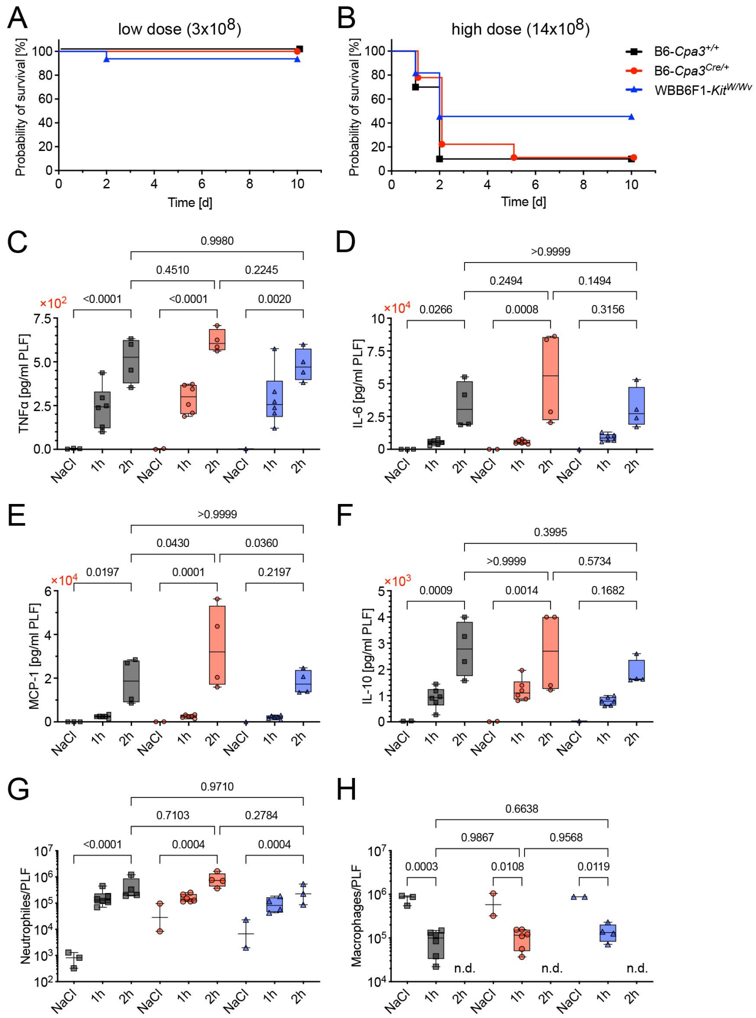

Figure 2.

Kit$^{W/Wv}$ mice are as resistant to injection of cecal slurry as *Kit*$^{+/+}$ mice.

(A, B). Survival of B6-*Cpa3*$^{+/+}$, B6-*Cpa3*$^{Cre/+}$, and WBB6F1-*Kit*$^{W/Wv}$ mice following injection of low dose (3 × 10^8) bacteria (A), or high dose (14 × 10^8) bacteria (B) from intestines of normal C57BL/6 mice. For both doses, the outcome for *Kit*$^{W/Wv}$ mice was statistically comparable to that of *Kit* wild-type mice, and mast cell deficiency (B6-*Cpa3+/+* versus B6-*Cpa3Cre/+*) also played no role in survival. Low dose survival: +/+ 100% of n = 12; Cre/+ 100% of n = 16; W/Wv 94% of n = 16 and p (+/+ vs W/Wv) = 0.3865; high dose survival: +/+ 10% of n = 10; Cre/+ 11% of n = 9; W/Wv 45% of n = 11 and p (+/+ vs W/Wv) = 0.1110. P-values of the survival curve comparisons were calculated using the Mantel-Cox Log-rank test. (C-F). Concentration and kinetic of inflammatory cytokine responses in peritoneal lavage fluid from B6-*Cpa3*$^{+/+}$, B6-*Cpa3*$^{Cre/+}$ and WBB6F1-*Kit*$^{W/Wv}$ after saline (NaCl) control injections, or one or two hours after low dose (3-6 × 108 bacteria) i.p. injection. Boxes in the plots extend from the 25th to 75th percentiles and the median is indicated. Whiskers range from minimum to maximum. Each dot represents an individual mouse. For p values, we compared the 2-hour anti-bacterial response vs saline for each genotype, and the response at 2 hours between the genotypes. P-values were calculated by ordinary one-way Anova with Šídák correction for multiple comparison. Sample numbers were for 1h: n = 6; 2h: n = 4 (for all genotypes); NaCl: n = 3 (+/+), n = 2 (*Cre/*+), n = 1 (*W/Wv*).

Absolute numbers of neutrophils (Gr1$^+$ CD11b$^+$) in the peritoneal cavity of B6-*Cpa3*$^{+/+}$, B6-*Cpa3*$^{Cre/+}$ and WBB6F1-*Kit*$^{W/Wv}$ after NaCl control injections, or one or two hours after low dose (3-6 × 10^8) bacteria i.p. injection. Group sizes were for NaCl: n = 3 (+/+), n = 2 (*Cre/*+), n = 2 (*W/Wv*); for 1h: n = 6 (+/+), n = 6 (*Cre/*+), and n = 4 (*W/Wv*), and for 2h n = 4 (+/+), n = 4 (*Cre/*+), and n = 3 (*W/Wv*). P-values were calculated on log-transformed values by ordinary one-way Anova with Šídák correction for multiple comparison.

(H). Absolute numbers of macrophages (Gr1$^-$ CD11b$^+$ F4/80$^+$) in the samples as in G. Sample size and calculation of P=values as in (G). 2h: not done (n.d.).

mice (donor flora) and injected them into recipient cohorts of $Cpa3^{+/+}$ (**Fig. 3A**), $Cpa3^{Cre/+}$; (**Fig. 3B**), or $Kit^{W/Wv}$ (**Fig. 3C**) mice (in this experiment all donor and recipient animals were on the WBB6F1 background). Cecal slurries with five different amounts of bacteria (2, 5, 10, 20, 40 × 108) were applied. This titration experiment revealed a clear correlation between bacterial load and mortality. Of note, we observed a marked leftward shift in the dose-response curves towards lower bacteria concentrations when injecting $Kit^{W/Wv}$ compared to $Kit^{+/+}$ (**Fig. 3A-C**). Collectively, these data indicate a higher pathogenicity of the microflora harbored in *Kit*-mutants compared to *Kit* wild-type mice.

To investigate whether the increased pathogenicity of $Kit^{W/Wv}$ microbiota is due to an altered bacterial composition of the cecal contents, we performed a basic microbiological examination and determined bacterial colony forming units (CFU) of the intestinal contents of $Kit^{+/+}$ and $Kit^{W/Wv}$ mice, including the specimens used for the i.p. infection experiments shown in **Fig. 3A-C**. To this end, we prepared serial dilutions of the cecal slurries, cultured these on blood agar and McConkey agar plates, and counted bacterial colonies. While Lactobacilli colonies were obtained at similar frequencies from isolates of both mouse strains, CFU counts for *Escherichia coli (E. coli*) were more that 1000 times higher in $Kit^{W/Wv}$ than in $Kit^{+/+}$ isolates (**Fig. 3D**). *Staphylococcus sp*. (most likely *Staph. Xylosus*) was sporadically found at low level in $Kit^{+/+}$ slurries (not shown). Hence, $Kit^{W/Wv}$ microbiota contains high levels of *E. coli*, which may underlie the observed pathogenicity.

Cohousing transfers sepsis resistance to $Kit^{W/Wv}$ mice

To independently prove that a pathogenic microbiota in $Kit^{W/Wv}$ mice is responsible for the increased CLP-susceptibility of this strain independent of mast cells, we performed co-housing experiments. $Kit^{W/Wv}$ and their wild-type $Kit^{+/+}$ littermates are F1 offspring from WB-$Kit^{W/+}$; and C57BL/6-$Kit^{Wv/+}$ parents. The phenotypically different *KitW/Wv* (white) and *Kit*+/+ (black) mice were normally separated at weaning, which may result in genotype-specific drifts of the enteric microbiota. To prevent such drift, we co-housed $Kit^{W/Wv}$ and $Kit^{+/+}$ mice, and repeated cecal ligation and puncture experiments. In contrast to the previous CLP experiments with non-co-housed mice at which severe (22 G needle) conditions were lethal for all $Kit^{W/Wv}$ mice (**Fig. 1B**), now with co-housed mice there remained no difference in survival (p = 0.3809) comparing $Kit^{W/Wv}$ (74 % of n = 38) and $Kit^{+/+}$ (65 % of n = 25) mice (**Fig. 3D**). These experiments suggest that co-housing transfers CLP-resistance from $Kit^{+/+}$ to $Kit^{W/Wv}$ mice.

Discussion

$Kit^{W/Wv}$ mice have higher mortality rates compared to $Kit^{+/+}$ control mice[1] in the clinically relevant cecal ligation and puncture sepsis model[24]. The absence of mast cells has been held responsible for this presumed deficiency of $Kit^{W/Wv}$ mice to mount an adequate anti-bacterial response which would ensue normal survival. In this study, we re-addressed the roles of mast cells and Kit in cecal ligation and puncture experiments. Clearly, absence of mast cells in *Kit* wild-type mice had no impact on mortality, hence, mast cells play no role in protection from cecal ligation and puncture sepsis. The cecal ligation and puncture data shown in this manuscript are based on experiments conducted over eight years by three independent experimentalists in two institutions (University of Ulm, Germany and German Cancer Research Center in Heidelberg, Germany). Altogether, the sample sizes (n's) are very large and we consider the data robust. Regarding Kit, we confirmed the susceptibility of $Kit^{W/Wv}$ mice to cecal ligation and puncture sepsis. Remarkably, $Kit^{W/Wv}$ mice were as protected as their wild-type controls when intestinal bacteria were injected (no puncture). Indeed, based on mortality, cytokine responses and neutrophil influx they were indistinguishable from $Kit^{+/+}$ controls. However, *KitW/Wv* mice differed from wild-type control mice in their microbiota. Based on in vivo lethality and in vitro colony formation, $Kit^{W/Wv}$ intestines contained more pathogenic bacteria than $Kit^{+/+}$ intestines. This suggests that in cecal

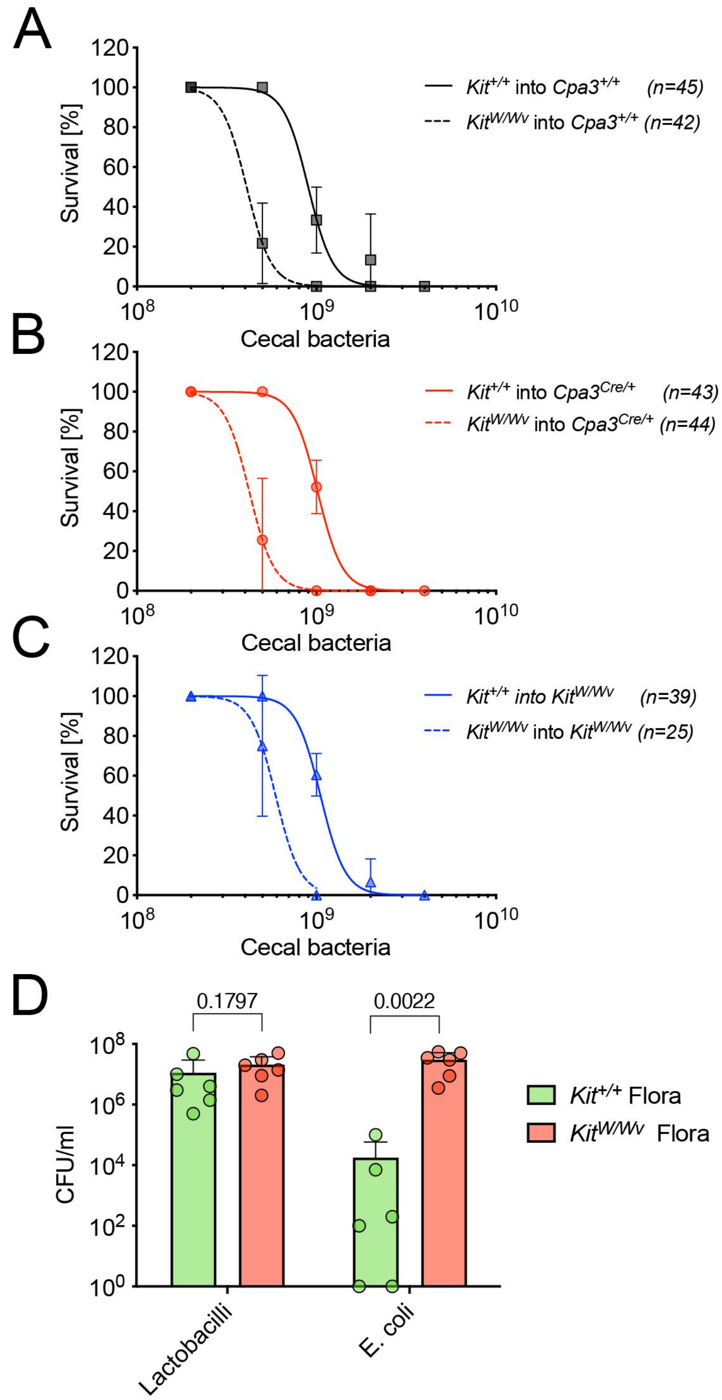

Figure 3.

Kit$^{W/Wv}$ mice harbor more pathogenic enteral bacteria than *Kit*$^{+/+}$ mice.

(A-C). Survival of *Cpa3*$^{+/+}$ (A), *Cpa3*$^{Cre/+}$ (B), and *Kit*$^{W/Wv}$ (C) mice following injection of increasing doses of bacteria (2, 5, 10, 20 or 40 × 10^8) isolated from the cecum of *Kit*$^{+/+}$ or *KitW/Wv* donor mice. All animals in this experiment were on the WBB6F1 strain background. Group sizes (n) are given in the figure. (D). Bacterial colony forming units were determined for cecal slurries of *Kit*$^{+/+}$ (n = 6) and *Kit*$^{W/Wv}$ mice (n = 6) which included all cecal slurries used in (A-C). Bars graphs show the mean + SD, each dot represents counts from one individual mouse. P-values were calculated with the Mann-Whithney test.

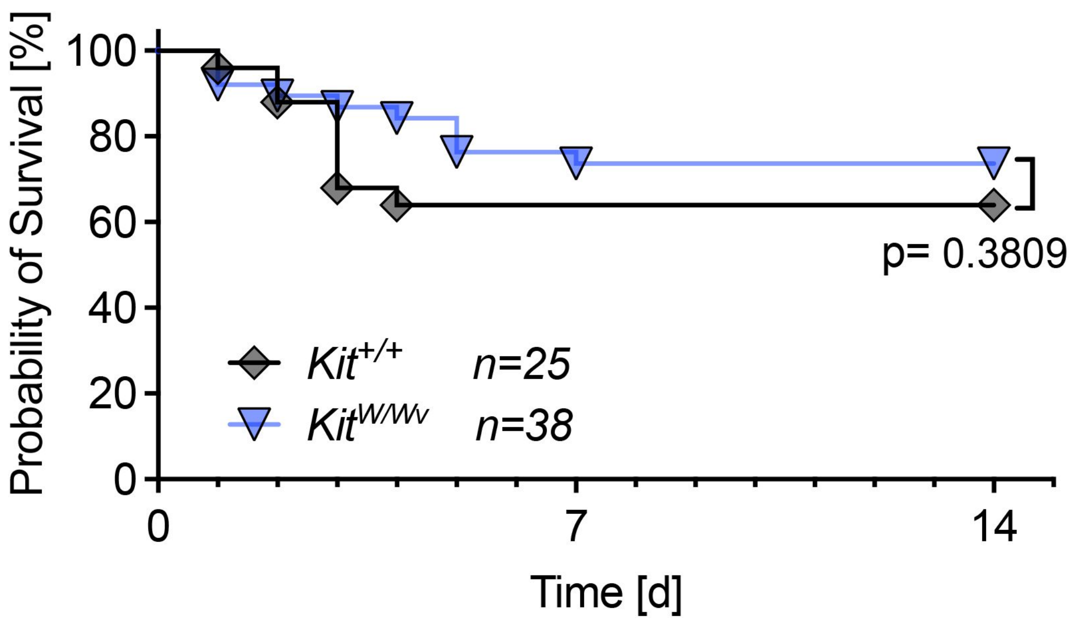

Figure 4.

Co-housing of *Kit*$^{W/Wv}$ with *Kit*$^{+/+}$ mice normalizes the susceptibility of KitW/Wv mice to cecal ligation and puncture.

Kit$^{+/+}$ and *Kit*$^{W/Wv}$ mice (WBB6F1 background) were housed together from weaning onwards until cecal ligation and puncture (22 G needle). This constant co-housing, increased the survival rate of *Kit*$^{W/Wv}$ mice (n=38) and equalized it (p=0.3809) to the probability of survival of *Kit*$^{+/+}$ mice (n=25). P-values were calculated using the Mantel-Cox Log-rank test.

ligation and puncture, $Kit^{W/Wv}$ mice release more pathogenic bacteria into their peritoneal cavity and are hence more severely challenged than $Kit^{+/+}$ mice. This would explain the stronger cytokine responses and the greater lethality of the *Kit*-mutant. In brief, the susceptibility of $Kit^{W/Wv}$ mice to cecal ligation and puncture is unrelated to mast cells or even 'immunology'. Our observation of incomparable intestinal microbiota comparing a mutant and its wild-type control may raise a note of caution on cecal ligation and puncture experiments comparing other mouse mutants and their wild-type counterparts.

There are conflicting reports regarding the role of mast cells in the cecal ligation and puncture assay, or in related bacterial infection models. Published results range from pro-pathogenic roles of mast cells[25] to enhanced vulnerablity of mouse mutants with loss of protease genes (Mcpt4; Mcpt6)[16,26]. It seems difficult to reconcile these reports with our experiments which show unimpaired survival of mast cell-deficient ($Cpa3^{Cre/+}$) mice. In a peritonitis model induced by injection of a mouse-virulent strain of *Klebsiella pneumoniae*, $Kit^{W/Wv}$ mice were reported to be impaired in bacterial clearance, neutrophil recruitment, and increased expression of TNFα in the peritoneal cavity[2]. In our hands, using the 'natural mixture' of cecal bacteria for intraperitoneal infections, $Kit^{W/Wv}$ mice showed comparable survival, TNFα secretion and neutrophil recruitment to the peritoneal cavity, all of which does not agree with the immunodeficiency that Malaviya et al. described[2]. The reasons for these discrepancies are again unclear. Naturally, host-to-host transmission of *Klebsiella pneumoniae* requires close contact and generally occurs through the fecal-oral route[27] rather than experimental intraperitoneal infections. Rather than probing immunity against single bacterial strains, such as *Klebsiella pneumoniae* or *E. coli*, we chose a more natural polymicrobial model, mimicking the release of enteral bacteria following gastrointestinal rupture. In such pathology, we found no evidence for a protective role of mast cells. This conclusion aligns with recent proteome data obtained from primary mouse and human mast cells[28]. By comparison with macrophages, ex vivo isolated mast cells lacked several different classes of pattern recognition receptors which suggests that peritoneal mast cells, at least at steady state, do not possess broad innate pattern recognition capabilities. Specifically, primary human and mouse mast cells do not express detectable amounts of TLRs, and their expression of MYD88 most likely does not represent a configuration for TLR but rather for IL-1R signaling. Mast cells share expression of the RNA sensor RIG-I and its downstream transcription factor IRF3 with most other immune cells, but do not express other RIG-I-like receptors or their downstream signaling components. Mast cells also do not express C-type lectin receptor components, or inflammasome components[28]. Importantly, TNFα protein, reported to be stored in mast cell secretory granules in vitro[29], was also undetectable in primary mast cells[28]. These data, together with our polymicrobial sepsis experiments, make it likely that mast cell play no role in protection against peritoneal sepsis. Instead, it appears that the principle innate responder cells in peritoneal sepsis are macrophages and neutrophils which are also the main cellular sources of TNFα.

The differences in composition and pathogenicity of *Kit*-mutant and wild-type enteral microbiota we report here can only directly refer to the mice bred and maintained during the experimental period in our mouse facilities. Breeding *Kit*+/+ and $Kit^{W/Wv}$ mice from WB-$Kit^{W/+}$ and B6-$Kit^{Wv/+}$; parents, and separating the offspring at the time of weaning, led to the observed differences in microbiota within the same mouse facility. Performing sepsis experiments in mice generated in this manner revealed that lack of Kit can strongly enhance the pathogenicity and, in particular, hugely increase numbers of *E. coli* colony-forming units. Our data therefore raise the possibility that published work comparing $Kit^{+/+}$ and $Kit^{W/Wv}$ mice may also have been based on separately kept mice. We cannot predict the outcome in microbiota composition of $Kit^{+/+}$ and $Kit^{W/Wv}$ mice bred elsewhere, but we suggest that investigators may control for this. Of particular importance in this regard appear to be co-housing experiments between *Kit*-mutants and wild-type mice.

Methods

Mice

B6-*Cpa3*$^{Cre/+}$ mice were obtained after backcrossing the *Cpa3*^{Cre} allele (*Cpa3*$^{tm3(icre)Hrr}$6 PMID: 22101159) for >20 generations onto C57BL/6J (B6). In all experiments, littermates from the breeding of B6-*Cpa3*$^{Cre/+}$ with wild-type B6 mice were used. *Kit*-mutant mast-cell-deficient WBB6F1-*Kit*$^{W/Wv}$ mice and their WBB6F1-*Kit*$^{+/+}$ littermate controls were obtained by crossing WB-*Kit*$^{W/+}$ with B6-*Kit*$^{Wv/+}$ mice (both originally obtained from Japan-SLC, Shizuoka, Japan). Unless otherwise described in the text, WBB6F1-*Kit*$^{W/Wv}$ mice and their WBB6F1-*Kit*$^{+/+}$ littermates were separated by coat color at weaning. For direct comparison, WBB6F1-*Cpa3*$^{Cre/+}$ and WBB6F1-*Cpa3*$^{+/+}$ littermates were generated in F1 intercrosses of WB-*Kit*$^{+/+}$ with B6-*Cpa3*$^{Cre/+}$ mice.

All animals were generated at the mouse breeding facilities of the University Clinics in Ulm or the center for preclinical research at the DKFZ in Heidelberg. All experiments were conducted in accordance to animal care guidelines pertaining to local animal committees (Regierungspräsidium Tübingen or Regierungspräsidium Karlsruhe) and to the institutional guidelines.

Sepsis models

Cecal ligation and puncture (CLP)

The surgical sepsis model was performed according to Rittirsch et al.[30] and Cuenca et al.[31]. In brief, mice were anesthetized (100 mg/kg ketamine and 16 mg/kg xylazine in saline, i.p.), belly shaved and placed onto a sterile-covered warming plate for the duration of the surgery. After disinfection with 70 % ethanol, the abdomen of the mice was opened with scissors by a 1-cm mid line cut into the skin, followed by a 1-cm incision of the peritoneum. The cecum was exteriorized and its stool content was gently palpated towards the distal end. The ligation with 4-0 sutures was placed at half the distance ('50 % ligation') between the distal pole and the base of the cecum. A single puncture was applied to the pole of the cecum (gauge sizes of the used needles are indicated for each experiment). Gentle pressure on the ligated cecum released a tiny drop of the cecal content while the needle was withdrawn to prevent immediate closure of the puncture hole. Afterwards the cecum was reposed and wound closure was performed with surgical clips. Sham control mice underwent the same surgical procedure, however without ligation and puncture of the cecum. Finally, mice received 1 ml saline s.c. for fluid replacement. For the entire observation period after surgery, all mice were closely monitored, and animals showing signs of severe distress or suffering were euthanized. Hence, statements in this manuscript referring to mortality all fell under this animal welfare protocol. For profiling of inflammatory cytokines, mice were sacrificed at the indicated time after surgery and blood for serum preparation was collected by heart puncture.

Cecal slurry injection

For the non-surgical sepsis model, suspensions of cecal bacteria were prepared from donor mice and injected intraperitoneally into the test mice. For preparation of the cecal slurry, four donor mice (B6-*Cpa3*$^{+/+}$; (**Fig. 2**), WBB6F1-*Kit*$^{+/+}$ or WBB6F1-*Kit*$^{W/Wv}$ (**Fig. 3**)) were sacrificed and the content of their cecum was pooled and resuspended in 10 ml saline (0.9 M NaCl). The suspension was centrifuged for 1 min at 20 g (200 rpm) in a swing-out bucket and the upper 8 ml were subsequently passed through 100 µm and 40 µm filters. Bacteria counts (rod-shaped) were determined using a Neubauer counting chamber with 0.02 mm chamber depth. Injected bacterial doses are indicated in **Figures 2** and **3**. Injected mice were closely monitored and moribund mice were sacrificed (see 'Cecal ligation and puncture' paragraph above). For the analysis of cytokine concentrations and myeloid cell infiltrations in the peritoneal cavity, mice were sacrificed at the indicated time after cecal slurry injection and the peritoneal fluid was collected. Therefore, 700-800 µl sterile FACS buffer (PBS with 5 % FCS) were injected intraperitoneally, and after a short abdominal massage, the peritoneal cavity was opened by a 5 mm incision to retrieve about 500 µl of the peritoneal exudate suspension with a pipette. A small aliquot was used to determine the concentration of cells and the total amount of cells was calculated from the total injected volume. The peritoneal exudate suspension was then separated into a cellular and a soluble fraction by centrifugation for 5 min with 500 g (2200 rpm). The cell pellet was resuspended in FACS buffer for subsequent flow cytometric analysis. The soluble fraction was centrifuged for another 5 min at 16200 g (13000 rpm) and the supernatant was snap frozen in liquid nitrogen for later cytokine measurement.

CFU determination

CFU of the cecal slurry suspensions used in the sepsis experiment was determined by plating serial dilutions (10-3 – 10-10 in 0.9 M NaCl) of the slurry stock suspension in replicates on blood agar and McConkey agar. *Lactobacillus* and *Staphylococcus* colonies were counted on blood agar plates, *Escherichia coli* colonies on McConkey agar plates.

Flow cytometric assays

Cytometric bead array (CBA)

Cytokine concentrations of serum samples or peritoneal lavages were determined in a bead-based multiplex cytometric assay (Cytometric Bead Array (CBA) Mouse Inflammation Kit, BD Biosciences). Samples (undiluted and 1:20 dilutions) and cytokine standards (10 ng – 5 pg serial dilutions) were incubated with mixtures of fluorescent beads coated with capture antibodies against IL-6, IL-10, INF-γ, MCP-1, TNF-α and IL12p70. PE-labeled detection reagent (i.e. PE-conjugated antibodies specific for the respective mouse cytokines) was added, and, after incubation and wash, samples were measured at a BD FACSCanto™ or BD LSRFortessa™ instrument. Cytokine concentrations were calculated from mean fluorescence values using the BD FCAP Array Software.

Peritoneal exudate cells (PEC)

The cellular fraction of the peritoneal exudate suspensions was analyzed for CD117+ mast cells, Gr1$^+$ CD11b$^+$ granulocytes and Gr1–CD11b$^+$ F4/80$^+$ macrophages on a BD FACS Canto. Therefore, cells were first incubated with mouse IgG (300 µg/ml, Dianova) for blocking of Fc-receptors and then stained with titrated amounts of fluorescent-labeled antibodies in PBS with 5 % FCS. The following antibodies were used: CD117-APC (2B8), CD11b-FITC (M1/70), Gr1-PE (RB6-8C5) all from Pharmingen, and F4/80-APC-A780 (BM8) from ebioscience.

Acknowledgements

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Reviewer #1 (Public review):

Summary:

Mast cells have previously been reported to play an important role in bacterial immune defense and act protectively in sepsis. However, many of these findings were based on studies using Kit mutant mice. In this study, the authors conducted a detailed investigation using mast cell-deficient Cpa3 Cre-Master mice. As a result, the authors found that the Cpa3 Cre-Master mice exhibited responses similar to wild-type mice in terms of bacterial immune defense. This suggests that the observed phenotype is not due to mast cell-dependent bacterial immune defense, but rather is associated with dysbiosis of the gut microbiota.

Strengths:

Mast cells have long been reported to play an important role in the protective response against sepsis, and their function in infection defense has been demonstrated. However, Kit mutant mice have been reported to exhibit impaired peristalsis, and several mast cell-specific genetically modified mouse lines have since been developed and examined in detail. This study presents an important finding by logically demonstrating that the exacerbation of sepsis in Kit mice is due to alterations in the gut microbiota, and that the phenotype previously thought to be mast cell-dependent was, in fact, not.

In addition, the experiments were carefully designed using mice with matched genetic backgrounds. These findings underscore the importance of microbiota composition in interpreting immune phenotypes and highlight the need for co-housing controls in mutant mouse studies.

A major strength of this work is the robustness of the CLP data, generated over eight years by three independent researchers across two institutions with large sample sizes, lending strong support to the conclusions.

Weaknesses:

The study assesses only a limited subset of gut bacterial species, leaving the extent to which E. coli expansion contributes to the observed phenotype unclear. Moreover, in the cohousing experiments, there is no evidence provided to confirm successful microbiota normalization between groups. A more detailed analysis of the microbial composition would be necessary to strengthen the reliability of the findings.

It is also important to note that Cpa3-deficient mice exhibit not only mast cell depletion but also defects in basophils and T cells. These additional immunological alterations may counterbalance one another, potentially masking phenotypic changes and complicating interpretation.

Furthermore, it remains to be determined whether the altered gut microbiota observed in KitW/Wv mice is a consequence of impaired intestinal motility, whether a similar phenotype is observed in KitW-sh/W-sh mice, and whether comparable results occur in SCF-deficient models. Addressing these questions would provide greater clarity on the contribution of mast cells versus secondary factors in the observed phenotypes.

Given that KitW/Wv mice exhibit impaired peristalsis, is the observed increase in E. coli a consequence of this dysfunction?

Previous studies with BMMC reconstitution experiments have indicated that mast cells are a source of TNF - how does this align with the current findings?

https://doi.org/10.7554/eLife.106789.1.sa2

Reviewer #2 (Public review):

Summary:

This study presents a useful finding that the high susceptibility to CLP sepsis of Kit-mutant mice is not due to mast cell deficiency, but to dysbiosis.

However, the present data are insufficient and incomplete to support the conclusion, and would benefit from more rigorous approaches. With the mechanism part strengthened, this paper would be of interest to researchers on mast cell biology and mucosal immunology.

Recommendations:

(1) The authors showed that E. coli increases in the cecum of Kit-mutant mice, which causes high CLP susceptibility. However, they did not provide any evidence E. coli is responsible for the high susceptibility. In the Figure 3 experiments, the authors administered the same number of cecal bacteria and did not show the number of E. coli after the administration. The authors should provide evidence showing that depletion of E. coli decreases susceptibility.

(2) The author should provide direct evidence of dysbiosis by, for example, shotgun sequencing of cecal and fecal contents.

(3) In case the authors find dysbiosis, they should analyze the mechanisms by which Kit mutation causes dysbiosis.

https://doi.org/10.7554/eLife.106789.1.sa1

Author response:

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *Mast cells have previously been reported to play an important role in bacterial immune defense and act protectively in sepsis. However, many of these findings were based on studies using Kit mutant mice. In this study, the authors conducted a detailed investigation using mast cell-deficient Cpa3 Cre-Master mice. As a result, the authors found that the Cpa3 Cre-Master mice exhibited responses similar to wild-type mice in terms of bacterial immune defense. This suggests that the observed phenotype is not due to mast cell-dependent bacterial immune defense, but rather is associated with dysbiosis of the gut microbiota.*
>
> *Strengths:*
>
> *Mast cells have long been reported to play an important role in the protective response against sepsis, and their function in infection defense has been demonstrated. However, Kit mutant mice have been reported to exhibit impaired peristalsis, and several mast cell-specific genetically modified mouse lines have since been developed and examined in detail. This study presents an important finding by logically demonstrating that the exacerbation of sepsis in Kit mice is due to alterations in the gut microbiota, and that the phenotype previously thought to be mast cell-dependent was, in fact, not.*
>
> *In addition, the experiments were carefully designed using mice with matched genetic backgrounds. These findings underscore the importance of microbiota composition in interpreting immune phenotypes and highlight the need for co-housing controls in mutant mouse studies.*
>
> *A major strength of this work is the robustness of the CLP data, generated over eight years by three independent researchers across two institutions with large sample sizes, lending strong support to the conclusions.*
>
> *Weaknesses:*
>
> *The study assesses only a limited subset of gut bacterial species, leaving the extent to which E. coli expansion contributes to the observed phenotype unclear.*

We will add new data based on 16S rRNA sequencing to the revised version.

> *Moreover, in the cohousing experiments, there is no evidence provided to confirm successful microbiota normalization between groups.*

We note that co-housing is a generally accepted method for microbiota equalization or conversion (Caruso et al., Cell Rep. 2019, Ridaura et al., Science 2013, and reviewed in Moore et al., Clin. Transl. Immunol. 2016). In any case, $Kit^{W/Wv}$ mutants were made resistant to CLP by co-housing. Similar microbiota sequencing results between groups,while useful, would again only be correlative.

> *A more detailed analysis of the microbial composition would be necessary to strengthen the reliability of the findings.*

See above

> *It is also important to note that Cpa3-deficient mice exhibit not only mast cell depletion but also defects in basophils and T cells. These additional immunological alterations may*

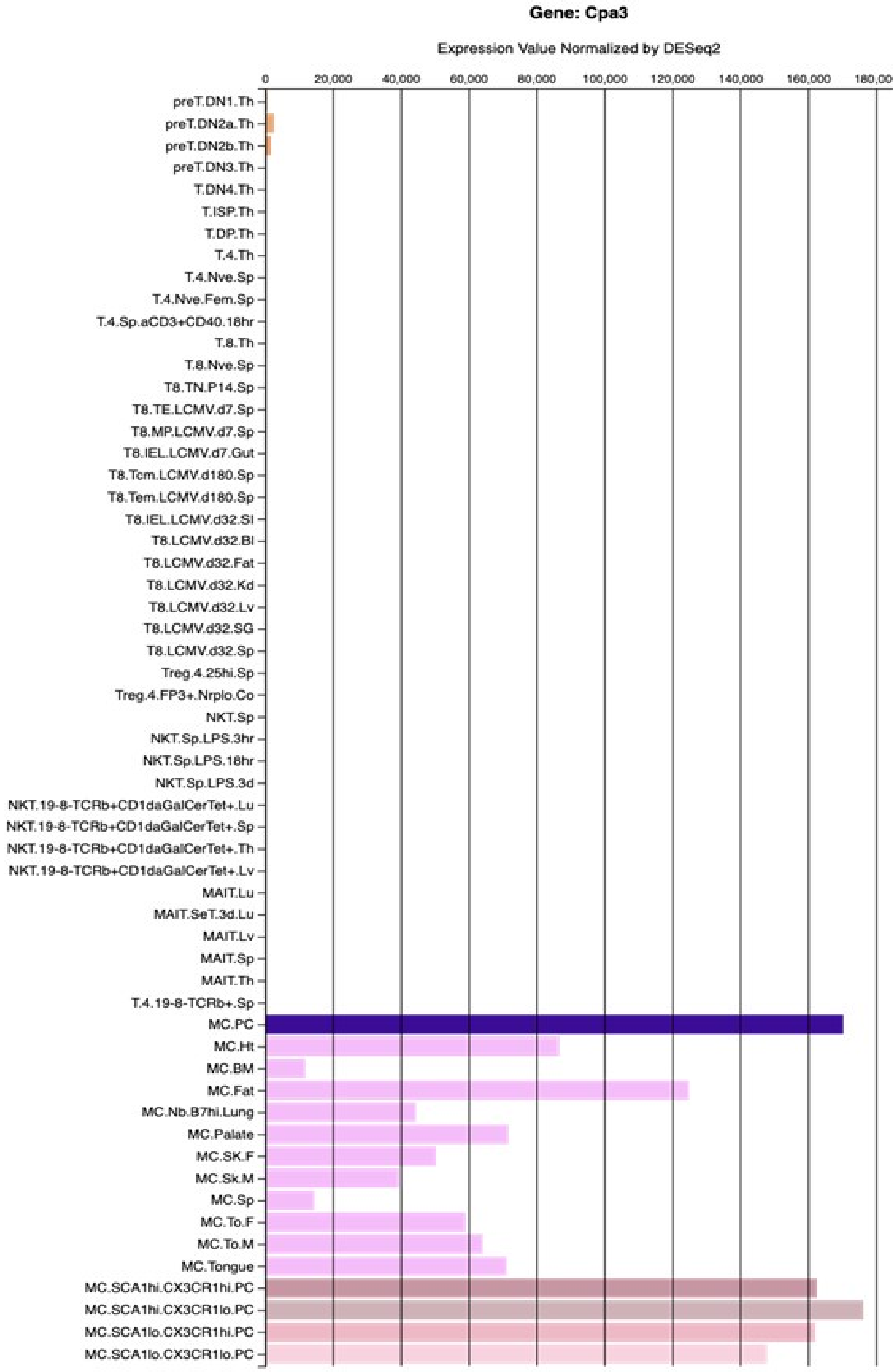
> *counterbalance one another, potentially masking phenotypic changes and complicating interpretation.*

Regarding basophils in *Cpa3*^{Cre} mice, compared to wild-type mice, basophils are reduced to about 39% of normal (Feyerabend et al., Immunity 2011). In *Kit*$^{W/Wv}$ mice, compared to wild-type mice, basophils are reduced to about 11% of normal. To our knowlegde, there has been no phenotype reported in which a reduction in basophils compensates for the loss for mast cells. Given that *Kit*$^{W/Wv}$ mice have about threefold lower numbers of basophils, and are highly susceptible to sepsis, there is no evidence that a reduction in basophils is protective in mast cell-deficient mice. On the contrary, mice that were normal for mast cells but had their basophils depleted were more susceptible to sepsis (Piliponsky et al., Nat. Immunol. 2019). Hence, basophils appear to be protective, and their reduction increases susceptibility. In light of these data and considerations, there is no evidence for a reduction in basophils to counterbalance the loss of mast cells in *Cpa3*^{Cre} mice.

Regarding T cells, there is no evidence, and there are no reports, that *Cpa3*^{Cre} mice have defects in T cells (Feyerabend et al., Immunity 2011, Feyerabend et al., Cell Metabolism 2016). Cpa3 is weakly and transiently expressed early in the T cell lineage (Feyerabend et al., Immunity 2009; for expression levels in T cells versus mast cells, see below figure from the Immgen Database). In summary, in contrast to the reviewer's claim, there are no known defects in T cell development or functions in *Cpa3*^{Cre} mice.

Author response image 1.

Generated from the Immgen database. Shown are RNAseq gene expression levels of diverse T-cell and mast cell populations.



Furthermore, it remains to be determined whether the altered gut microbiota observed in Kit$^{W/Wv}$ mice is a consequence of impaired intestinal motility, whether a similar

> *phenotype is observed in KitW-sh/W-sh mice, and whether comparable results occur in SCF-deficient models. Addressing these questions would provide greater clarity on the contribution of mast cells versus secondary factors in the observed phenotypes.*

Mice without mast cells (*Cpa3*^{Cre} mice) are as resistant to sepsis as wild-type mice. Hence, mast cells are not involved in the immunity against sepsis, and 'secondary factors' are not involved in this simple experiment (both groups of mice, wild type and *Cpa3*^{Cre} mice, were on the idential genetic background). Second, *Kit*$^{W/Wv}$ mice are also as resistant to sepsis as wild-type mice when confronted with the identical intestinal slurry. Therefore, *Kit*$^{W/Wv}$ mice have no immune deficit in response to sepsis. Hence, in our view, the underlying immunological question regarding the role of mast cells in sepsis has been conclusively addressed by our data. Future studies may address the mechanism that causes dysbiosis in *Kit*$^{W/Wv}$ mice, and other Kit mutants and steel mutants could be examined as well. These questions are, however, unrelated to the role of mast cells in sepsis, or the response of *Kit*$^{W/Wv}$ mice to sepsis, and would therefore not affect the central conclusion of our manuscript ("Susceptibility of Kit-mutant mice to sepsis caused by enteral dysbiosis, not mast cell deficiency").

> *Given that Kit*$^{W/Wv}$ *mice exhibit impaired peristalsis, is the observed increase in E. coli a consequence of this dysfunction?*

See above

> *Previous studies with BMMC reconstitution experiments have indicated that mast cells are a source of TNF - how does this align with the current findings?*

It is possible that cultured and transplanted mast cells (BMMC) produce TNF. Given that we did not find a reduction in TNF levels in the peritoneal lavage or serum in mice without mast cells undergoing sepsis, under physiological conditions mast cell-derived TNF does not seem to have a measuable impact on total TNF levels.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *This study presents a useful finding that the high susceptibility to CLP sepsis of Kit-mutant mice is not due to mast cell deficiency, but to dysbiosis.*
>
> *However, the present data are insufficient and incomplete to support the conclusion, and would benefit from more rigorous approaches. With the mechanism part strengthened, this paper would be of interest to researchers on mast cell biology and mucosal immunology.*

We disagree with this view that our data are insufficient and incomplete. Our results demonstrate that mice lacking mast cells (*Cpa3*^{Cre} mice) are as resistant to sepsis as wild-type mice, indicating that mast cells do not play a detectable role in immunity against sepsis. Additionally, we show that *Kit*$^{W/Wv}$ mice exhibit the same resistance to sepsis as wild-type mice when confronted with the identical intestinal slurry. This finding demonstrates that *Kit*$^{W/Wv}$ mice have no immune deficit in response to sepsis. These central data are both sufficient and complete, given that our data fully address the immunological questions regarding the role of mast cells in sepsis. Our study aimed to investigate the role of mast cells in sepsis, not to examine the mechanisms of dysbiosis or associated pathological phenotypes in Kit mutant controls.

> *Recommendations:*

> *(1) The authors showed that E. coli increases in the cecum of Kit-mutant mice, which causes high CLP susceptibility. However, they did not provide any evidence E. coli is responsible for the high susceptibility.*

We showed that E. coli CFUs were increased in the cecum of Kit-mutant mice, but we did not state that this causes CLP susceptibility. We wrote: 'Hence, $Kit^{W/Wv}$ microbiota contains high levels of E. coli, which may underlie the observed pathogenicity'. We demonstrated that intestinal slurry from $Kit^{W/Wv}$ mice is more pathogenic compared to intestinal slurry from wild-type mice. However, we did not search for, or identify the bacterial species that causes this increased pathogenicity because we were adressing the role of mast cell in sepsis.

> *In the Figure 3 experiments, the authors administered the same number of cecal bacteria and did not show the number of E. coli after the administration.*

The samples were split and one aliquot was analysed by microbiology and the other aliquot was injected intraperitoneally. Fig. 3d shows the colony forming units (for Lactobacilli and E coli) from aliquots of cecal slurry used in the intraperitoneal injection experiments shown in Fig. 3a-c. Hence, our data show the colony forming units that were injected into the mice. It is unclear to us why this is not the key information rather than 'the number of E. coli after the administration'.

> *The authors should provide evidence showing that depletion of E. coli decreases susceptibility.*

See response to point 1 above.

> *(2) The author should provide direct evidence of dysbiosis by, for example, shotgun sequencing of cecal and fecal contents.*

The large increase in E coli counts in $Kit^{W/Wv}$ is evidence of dysbiosis. To obtain data beyond classical microbiology, we also performed 16S rRNA sequencing which will be included in the revision.

> *(3) In case the authors find dysbiosis, they should analyze the mechanisms by which Kit mutation causes dysbiosis.*

The mechanism that causes dysbiosis in $Kit^{W/Wv}$ mice (which emerged from our work) belongs to other research areas that address the role of Kit in intestinal pathophysiology. These questions are unrelated to the role of mast cells in sepsis, or the response of $Kit^{W/Wv}$ mice to sepsis. Regardless of the results of such experiments, the conclusion ("Susceptibility of Kit-mutant mice to sepsis caused by enteral dysbiosis, not mast cell deficiency") remains unaffected. In brief, further explorations of pathological phenotypes of a control mutant will not add to the core message. Along these lines, the review process and the revision shall center on making the core of a paper as conclusive as possible, and not widen a paper by requests 'tangential to the main conclusion' (Kaelin Jr. Nature 2017).

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